Sorting

04 April 2026 16:31

Sorting → [1, 0, 3, 2, 7, 6]

→ [0, 1, 2, 3, 6, 7]

→ [7, 6, 3, 2, 1, 0]

Types of Sorting

① Bubble Sort.

[1, 0, 3, 2, 7, 6]

→ [0, 1, 3, 2, 7, 6]

→ [0, 1, 2, 3, 7, 6]

→ [0, 1, 2, 3, 6, 7]

Smaller than the first element

[6, 7, 1, 2, 0, 4]

[6, 1, 7, 2, 0, 4]

[6, 1, 2, 7, 0, 4]

[6, 1, 2, 0, 7, 4]

1st Iteration → [6, 1, 2, 0, 4, 7]

2nd → [1, 6, 2, 0, 4, 7] → Sorted

[1, 2, 6, 0, 4, 7]

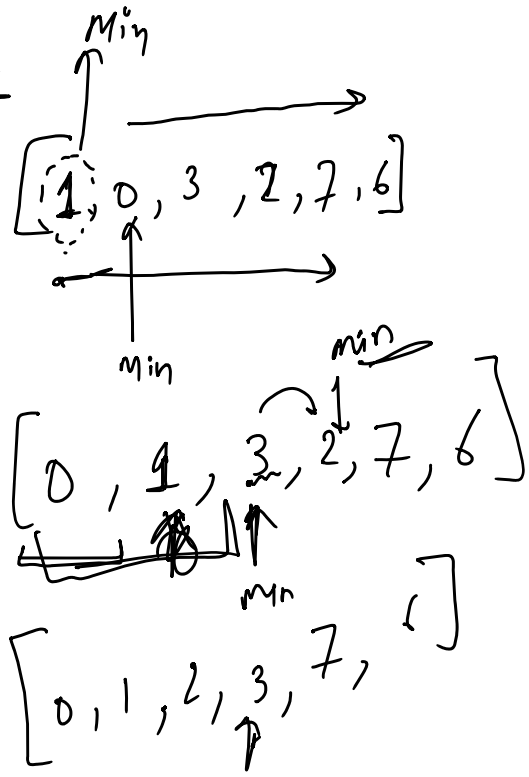
[1, 2, 0, 6, 4, 7]

arr = [1, 2, 0, 4, 6, 7] → Sorted.

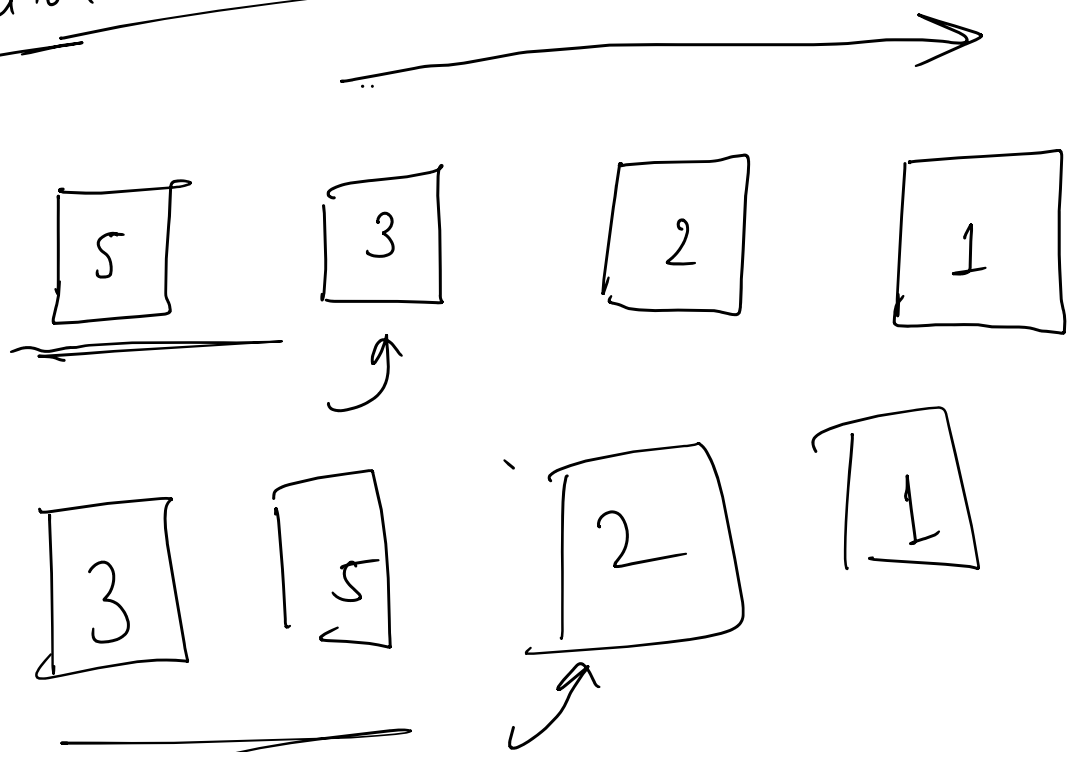
```
for (int i = 0 ; i < arr.length ; i++) {
    for (int j = 0 ; j < n-i-1 ; j++) {
        if (arr[j] > arr[j+1]) {
            // swap arr[j], arr[j+1]
```

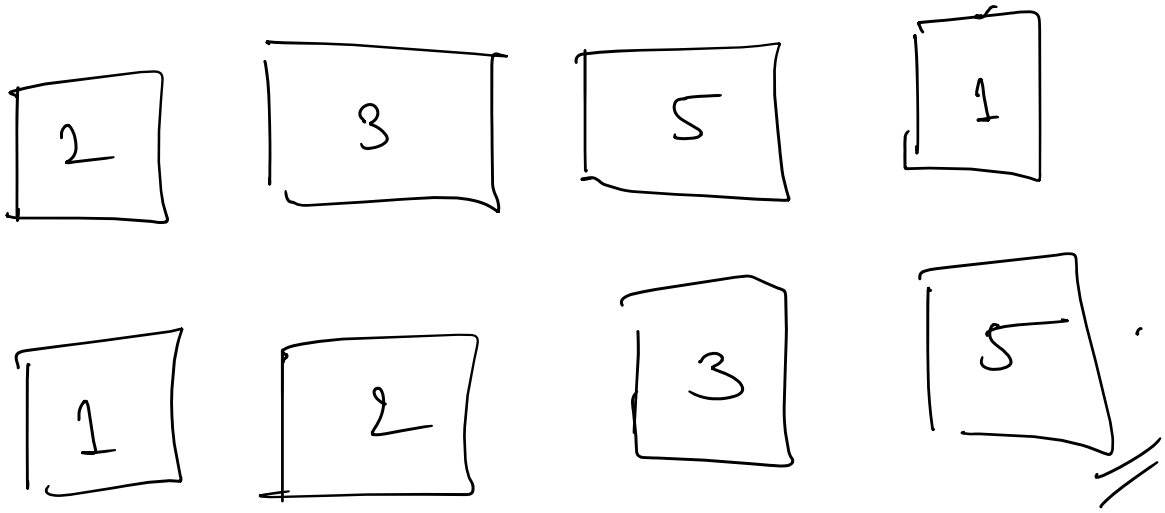
```
        // swap arr[j], arr[j+1]
    }
}
```

(2) Selection Sort



(3) Insertion Sort





[1, 0, 3, 2, 7, 6]

[0, 1, 3, 2, 7, 6]

[0, 1, 2, 3, 7, 6]

[0, 1, 2, 3, 6, 7]